

MTH 377/577 Convex Optimization

Practice Quiz

Instructions: Answer all questions.

Time: 60 minutes.

Maximum Marks: 20

1. State whether the following statements are true/false. For true statements, provide a short proof. For false, provide a counter example.

- (a) The set

$$A = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid \sum_{i=1}^n x_i \leq n\}$$

is compact. (3)

- (b) Consider a set

$$A = \{\theta_1 x_1 + \theta_2 x_2 \mid x_1 = (4, 5), x_2 = (1, 3), \theta_1, \theta_2 \geq 0\}.$$

A is a convex cone. (2)

- (c) The function

$$f(x) = -(x - 2)^2$$

is convex. (2)

- (d) The set

$$C = A \cup B$$

where A, B are convex, is always convex. (2)

Solution:

- (a) **False.**

Let

$$A = \left\{ (x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid \sum_{p=1}^n x_p \leq n \right\} \subset \mathbb{R}^n.$$

We show that A is *not compact*.

A subset of $\mathbb{R}^n$ is compact if and only if it is both closed and bounded .

Is A closed? Yes.

Let $f : X \rightarrow Y$ be a function and let $F \subset Y$ be a closed set. Then f is continuous if and only if $f^{-1}(F)$ is closed in X .

Note that for any $n \in \mathbb{N}$, the interval $(-\infty, n]$ is closed in $\mathbb{R}$. Define

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad f(x_1, x_2, \dots, x_n) = \sum_{p=1}^n x_p.$$

The function f is continuous.

Therefore,

$$A = f^{-1}((-\infty, n])$$

is closed in $\mathbb{R}^n$.

$$\boxed{A \text{ is closed in } \mathbb{R}^n.}$$

Is A bounded? No.

Pick an element of A of the form

$$(-t, 0, \dots, 0), \quad t > 0.$$

(Note: the negative sign is chosen to ensure that the sum remains $\leq n$.)

Then,

$$\|(-t, 0, \dots, 0)\| = t \xrightarrow[t \rightarrow \infty]{} \infty.$$

Hence, there does not exist any $K > 0$ such that

$$\|(-t, 0, \dots, 0)\| \leq K \quad \text{for all } t > 0.$$

$$\boxed{A \text{ is not bounded in } \mathbb{R}^n.}$$

Since A is closed but not bounded, it is not compact.

$$\boxed{A \text{ is not compact in } \mathbb{R}^n.}$$

(b) **True.**

Let

$$A = \{\theta_1 x_1 + \theta_2 x_2 : x_1 = (4, 5), x_2 = (1, 3), \theta_1, \theta_2 \geq 0\} \subset \mathbb{R}^2.$$

Cone property:

Let $x \in A$ and $\lambda \geq 0$. Then

$$x = \theta_1 x_1 + \theta_2 x_2 \quad \Rightarrow \quad \lambda x = (\lambda \theta_1) x_1 + (\lambda \theta_2) x_2 \in A.$$

Hence, A is a cone.

Convexity:

Let $x, y \in A$ and $\alpha \in [0, 1]$. Write

$$x = \theta_1 x_1 + \theta_2 x_2, \quad y = \gamma_1 x_1 + \gamma_2 x_2,$$

where $\theta_1, \theta_2, \gamma_1, \gamma_2 \geq 0$. Then

$$\alpha x + (1 - \alpha)y = (\alpha\theta_1 + (1 - \alpha)\gamma_1)x_1 + (\alpha\theta_2 + (1 - \alpha)\gamma_2)x_2 \in A.$$

Thus A is convex.

Hence, A is a convex cone.

(c) **False**

Let

$$f(x) = -(x - 2)^2.$$

We claim that f is *not convex*. For any $x_1, x_2 \in \text{dom} f$, $\theta \in [0, 1]$

$$f(\theta x_1 + (1 - \theta)x_2) = -[\theta x_1 + (1 - \theta)x_2 - 2]^2$$

$$\theta f(x_1) + (1 - \theta)f(x_2) = -\theta(x_1 - 2)^2 - (1 - \theta)(x_2 - 2)^2$$

Show that $f(\theta x_1 + (1 - \theta)x_2) < \theta f(x_1) + (1 - \theta)f(x_2)$. Alternatively, since f is twice differentiable on $\mathbb{R}$, we can use the **second derivative test to check convexity**.

Compute:

$$f'(x) = -2(x - 2), \quad f''(x) = -2.$$

Since

$$f''(x) = -2 < 0 \quad \forall x \in \mathbb{R},$$

the function f is concave and hence not convex.

f is not a convex function.

(d) **False**

Let A, B be convex sets. Then $A \cup B$ may or may not be convex.

Example (i):

Let

$$A = [1, 2], \quad B = (2, 3).$$

Then

$$A \cup B = [1, 3),$$

which is an interval in $\mathbb{R}$ and hence convex.

$$\boxed{A \cup B \text{ is convex.}}$$

Example (ii):

Let

$$A = [1, 2], \quad B = [3, 4].$$

Then

$$A \cup B = [1, 2] \cup [3, 4],$$

which is not an interval.

Since **a subset** $S \subset \mathbb{R}$ **is convex if and only if it is an interval**, we conclude that

$$\boxed{A \cup B \text{ is not convex.}}$$

Example (iii):

Let

$$A = \{(x, 0) \in \mathbb{R}^2 \mid x \geq 0\}, \quad B = \{(0, y) \in \mathbb{R}^2 \mid y \geq 0\}.$$

Here, A is the non-negative x -axis and B is the non-negative y -axis. Both A and B are convex subsets of $\mathbb{R}^2$.

However,

$$\frac{1}{2} \left(\frac{1}{2}, 0 \right) + \frac{1}{2} \left(0, \frac{1}{2} \right) = \left(\frac{1}{4}, \frac{1}{4} \right) \notin A \cup B.$$

Therefore,

$$\boxed{A \cup B \text{ is not convex.}}$$

2. Let $\{x_k\}_{k \in \mathbb{I}_+}$ be a sequence in $\mathbb{R}$ where

$$x_{k+1} = (-1)^k x_k.$$

Let X be the set of all terms x_k in the given sequence. Let

$$Y = \{x \in \mathbb{R} \mid x > 1\}.$$

- (a) Does there exist a hyperplane that separates X and Y ? If yes, find such a hyperplane. If not, why? Use separating hyperplane theorem(s) in support of your answer. (2.5)
- (b) Let $f : X \rightarrow \mathbb{R}$ and $g : Y \rightarrow \mathbb{R}$ be continuous functions. Do f and g have a maxima/minima? Provide an explanation for your answer. (2.5)

Solution: (a) Let $\{x_k\}_{k \in \mathbb{I}_+}$ be a sequence in $\mathbb{R}$, defined by the relation

$$x_{k+1} = (-1)^k x_k.$$

Take $x_1 = a \in \mathbb{R}$.

Then,

$$x_2 = -a, \quad x_3 = (-1)^2 x_2 = -a, \quad x_4 = (-1)^3 x_3 = a, \quad \dots$$

Therefore, the range of the above sequence is

$$\{-a, a\}.$$

Let

$$X := \text{the set of all terms of } \{x_k\}_{k \in \mathbb{I}_+} \quad (\text{range of the sequence}).$$

Define

$$Y := \{x \in \mathbb{R} \mid x > 1\}.$$

Thus,

$$X = \{-a, a\}, \quad a \in \mathbb{R},$$

and

$$Y = (1, \infty).$$

Note that $X, Y \subset \mathbb{R}$.

In $\mathbb{R}$, a hyperplane is of the form

$$H = \{x \in \mathbb{R} \mid hx = \beta\}, \quad \text{with } h \neq 0.$$

Case (i): $|a| < 1$

(Note that the comparison is with 1 since $Y = (1, \infty)$.)

Then,

$$\max X = |a| < 1.$$

Hence,

$$\max X \leq \inf Y.$$

Therefore, X and Y are disjoint, and Y is convex. In particular,

$$X \cap Y = \emptyset.$$

By the **weak separating hyperplane theorem (Lecture 6)**, there exists a hyperplane (h, β) such that

$$h^T x \leq \beta \leq h^T y \quad \forall x \in X, \forall y \in Y.$$

Since we are working in $\mathbb{R}$, choose

$$h = 1, \quad \beta = 1.$$

Then,

$$x \leq 1 \leq y \quad \forall x \in X, \forall y > 1.$$

Thus, the hyperplane $x = 1$ separates X from Y .

Case (ii): $|a| > 1$

Then either $a > 1$ or $a < -1$.

In either case,

$$\max\{a, -a\} > 1.$$

Hence,

$$X \cap Y \neq \emptyset.$$

Therefore, no separating hyperplane exists in this case.

(b) Let $f : X \rightarrow \mathbb{R}$ and $g : Y \rightarrow \mathbb{R}$ be continuous functions, where

$$X = \{-a, a\}, \quad Y = (1, \infty).$$

(i) Since X is a finite set, it is closed and bounded in $\mathbb{R}$. Hence, X is compact.

By Weierstrass theorem, the continuous function

$$f : X \rightarrow \mathbb{R}$$

attains both a maximum and a minimum on X .

(ii) The set Y is not compact. Therefore, the Weierstrass theorem does not apply.

Hence, the function

$$g : Y \rightarrow \mathbb{R}$$

need not attain a maximum or a minimum on Y .

For example, define

$$g : Y \rightarrow \mathbb{R}, \quad g(x) = x.$$

Then g has no minimum on Y . Moreover,

$$g(x) \rightarrow \infty \quad \text{as } x \rightarrow \infty,$$

and hence g has no maximum on Y .

3. Find the (i) convex hull, (ii) affine hull for the following set:

$$A = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}. (3)$$

Solution:

Let

$$x_1 = (1, 0, 0), \quad x_2 = (0, 1, 0), \quad x_3 = (0, 0, 1).$$

(i) **Convex hull**

Definition: The convex hull of A is

$$\text{conv}(A) = \{\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 \mid \lambda_1, \lambda_2, \lambda_3 \geq 0, \lambda_1 + \lambda_2 + \lambda_3 = 1\}.$$

Putting in the points,

$$\lambda_1(1, 0, 0) + \lambda_2(0, 1, 0) + \lambda_3(0, 0, 1) = (\lambda_1, \lambda_2, \lambda_3).$$

Thus,

$$\text{conv}(A) = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_i \geq 0, x_1 + x_2 + x_3 = 1\}.$$

$$\boxed{\text{conv}(A) = \{x \in \mathbb{R}^3 : x_i \geq 0, x_1 + x_2 + x_3 = 1\}}$$

(ii) **Affine hull**

Definition:

$$\text{affine}(A) = \{\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 \mid \alpha_1 + \alpha_2 + \alpha_3 = 1, \alpha_i \in \mathbb{R}\}.$$

Substituting,

$$\alpha_1(1, 0, 0) + \alpha_2(0, 1, 0) + \alpha_3(0, 0, 1) = (\alpha_1, \alpha_2, \alpha_3).$$

Hence,

$$\text{affine}(A) = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 1\}.$$

$$\boxed{\text{affine}(A) = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1\}}$$

4. Use Farkas' lemma and find out if the system $Ax = b$ has a non-negative solution when

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad x \in \mathbb{R}^3.$$

Show all the steps.

(3)

Solution:

Farkas' Lemma

Exactly one of the following systems has a solution:

- (a) $Ax = b, x \geq 0.$
- (b) There exists y such that

$$y^T A \geq 0, \quad y^T b < 0.$$

The Farkas' alternative is:

$$y_1 + 2y_2 + y_3 \geq 0$$

$$y_1 + 3y_2 \geq 0$$

$$y_1 + 2y_3 \geq 0$$

and

$$y_1 + 2y_2 + y_3 < 0$$

Note that the first inequality contradicts the last. Therefore, y does not exist. By Farkas Lemma, a non-negative solution for $Ax = b$ exists.